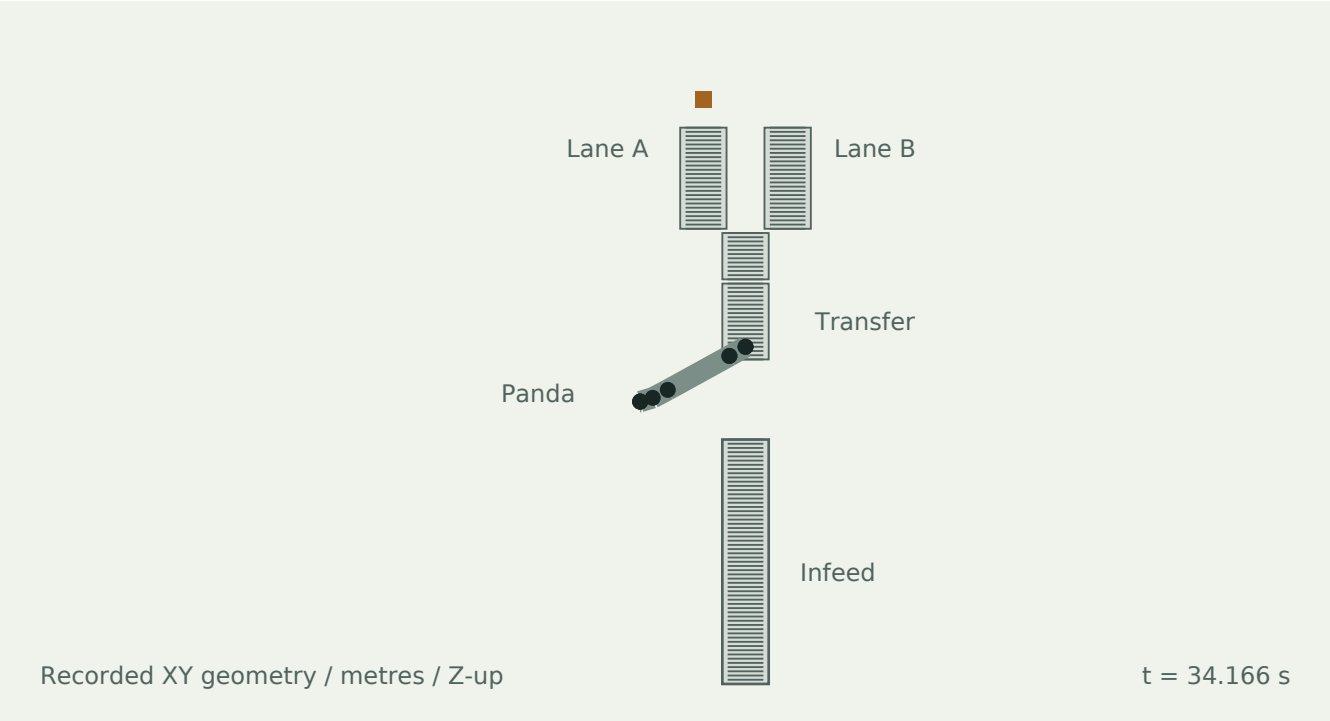
Carton sorting study

Grip force comparison

13 October 2026 / 18:33 UTC



MuJoCo Menagerie Franka Panda / Contact-driven sorting cell

Project: Unassigned / Creator: Not recorded

Executive finding

3 of 6 recorded trials completed the full sorting task. Training selection: baseline. Held-out trials are reported separately.

Physics figures use the first recorded trial, identified on each analysis page. This choice does not filter for success.

The controller used simulated object pose and mass, joint-space planning, position actuation and native frictional contacts. No hardware performance is inferred.

Conditions and measured results

Parameter	Saved value
Robot / engine	Franka Panda / MuJoCo 3.15.0
World version	vivacity.sorting-cell/1.0
Seed / timestep	19 / 0.002 s
Cartons / reference mass	1 / 0.060 kg
Box dimensions	0.050 m on each axis
Friction / gripper force scale	0.8 / 1.0
Conveyor / minimum arrival spacing	0.090 m/s / 18.000 s
Lane criterion	A below 0.07 kg; B otherwise
Initial carton X / Y	0.500 m / -0.400 m
Status / failure category	completed / None recorded
Verified deliveries / configured batch	1 / 1
Simulation duration / steps	34.166 s / 17083
Wall execution time	32.277 s
Delivered throughput	1.756 cartons/min

Interpretation

Throughput divides verified deliveries by elapsed simulation time. This finite batch is not a steady-state industrial throughput or a hardware cycle-time measurement. A carton counts only after native contact with its mass-selected bin floor.

Gravity is 9.81 m/s²; contacts use MuJoCo friction and compliant constraints. Robot arm gravity compensation is enabled. Object dimensions, friction and mass are configured model inputs, not camera estimates.

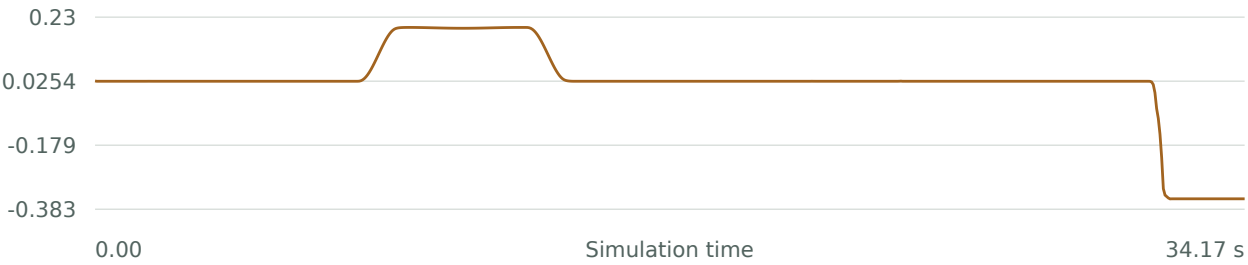
02 / PHYSICS ANALYSIS

Motion and contact

Source run d8544da0-508c-4d58-850c-e54729495e78 / 695 stored samples

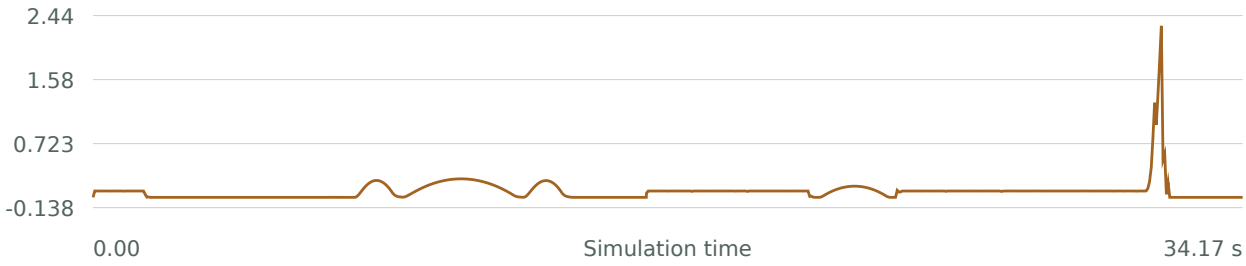
Carton height

m



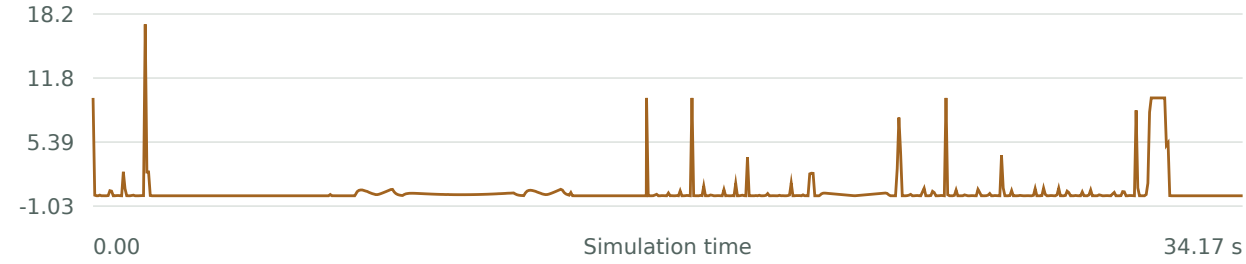
Carton speed

m/s



Carton acceleration magnitude

m/s²



Joint 1 actuator torque

N m



Pose time is shown. Forces and acceleration refer to the preceding 2 ms physics solve. Work is accumulated every integration step; other plotted quantities are sampled. Gaps mean unavailable data.

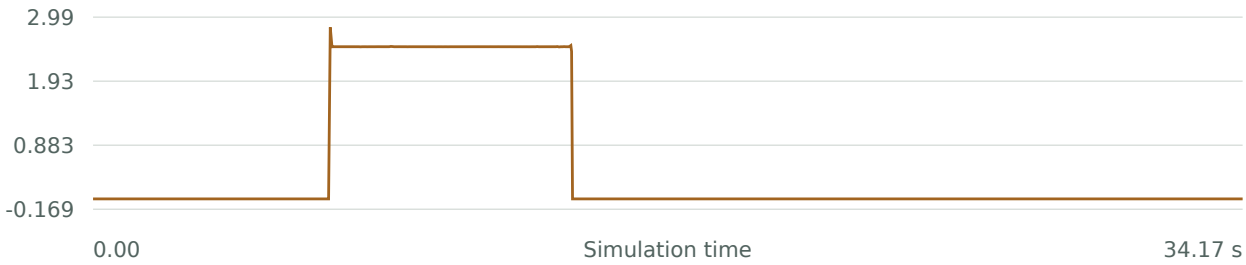
03 / PHYSICS ANALYSIS

Force and mechanical work

Source run d8544da0-508c-4d58-850c-e54729495e78 / 695 stored samples

Gripper contact normal force

N



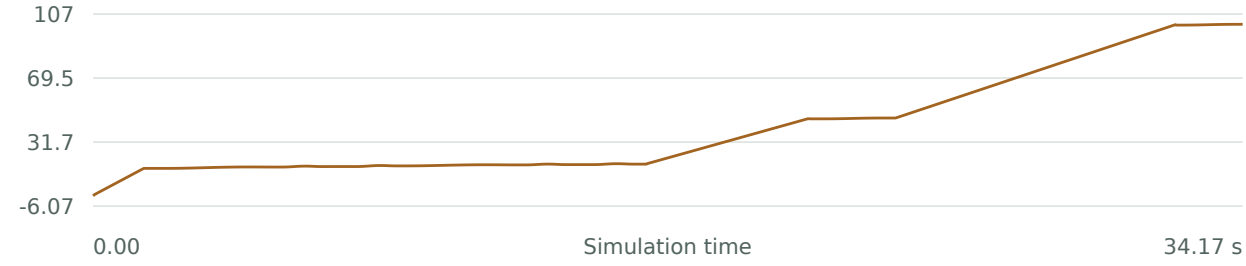
Total actuator mechanical power

W



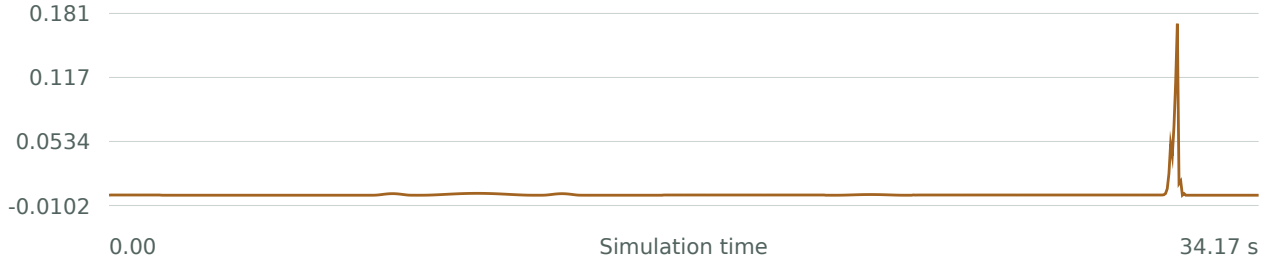
Net actuator mechanical work

J



Carton kinetic energy

J

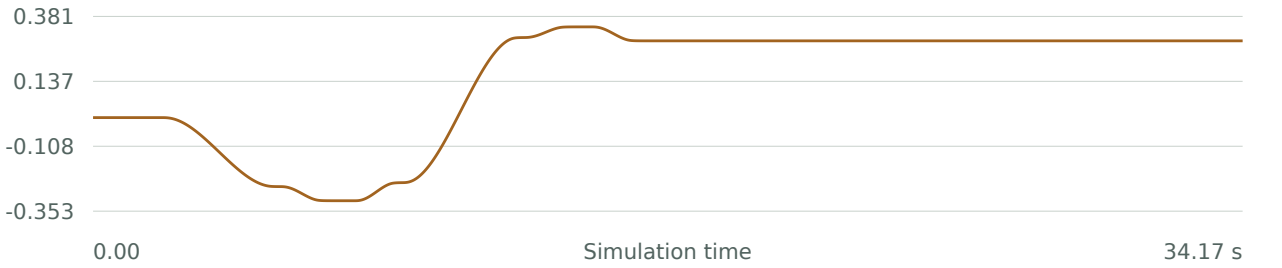


Pose time is shown. Forces and acceleration refer to the preceding 2 ms physics solve. Work is accumulated every integration step; other plotted quantities are sampled. Gaps mean unavailable data.

The recorded operation

Joint 1 position

rad



Simulation time	Recorded event
0.000 s	infeed started
2.046 s	station arrival
2.046 s	planning started
2.046 s	plan selected
5.560 s	approach completed
6.976 s	descend completed
7.776 s	close completed
9.188 s	lift completed
12.800 s	transport completed
14.222 s	lower completed
14.822 s	release completed
16.442 s	retract completed
16.442 s	physical transfer verified
21.574 s	route selected
32.166 s	destination arrival

05 / CONTROLLED COMPARISON

Repeated trials and held-out evidence

Every configured carton has bilateral grip, retained lift/transfer, release onto transfer rollers and contact with its mass-selected bin floor; no prohibited robot contact, controller, joint-limit or timeout failure.

Seed offset 100003; matched unseen starting X/Y perturbations up to 15/35 mm, clipped to the supported infeed. Settings frozen before any trial.

Condition / partition	Success	Mean duration	Sample variance
baseline / training	2/2	34.166 s	0.000 s²
candidate / training	0/2	Unavailable	Unavailable
baseline / held out	1/1	34.176 s	Unavailable
candidate / held out	0/1	Unavailable	Unavailable

Higher training success rate, then shorter mean successful simulation duration.

Trial outcomes

Trial / seed	Condition	Outcome
1 / 19	training / baseline	completed
2 / 19	training / candidate	lift_failure
3 / 20	training / baseline	completed
4 / 20	training / candidate	lift_failure
5 / 100022	held out / baseline	completed
6 / 100022	held out / candidate	lift_failure

Small seeded simulation sample. Held-out trials do not change selection. No hardware reliability or statistical significance is inferred.

APPENDIX / REPRODUCIBILITY

Evidence and physical assumptions

Saved record: 32d0e1aa-d544-4e44-af9f-85e89fa7af4b

Record SHA-256: 2037d3bd6ea6a79b7b0c2246beeb3d4cb91edcf18f0baab67761a752d08362e6

Native initial model SHA-256: e14b2c854cf1e6f6a1b1ee198955ce5df200b1e801bfea1628467a9135f4d2ee

Full integration state, controls, configuration, events and all stored samples are included in the JSON archive. The ZIP also contains a flat CSV of measured samples. No random data is generated during reporting.

cartons_sorted — Count of cartons with verified contact with their mass-selected bin floor.

success_rate — Completed full-batch trials / all trials in the named condition and partition.

simulation_time_s — Elapsed MuJoCo integration time, not wall time.

mechanical_work_j — Integral of summed actuator force times actuator velocity at every physics step.

missing — null means unavailable, never zero-filled. Electrical energy is not modeled.

Equations

Actuator mechanical power P = sum(force × actuator velocity). Net work W = sum(P × 0.002 s). Kinetic energy includes translation and rotation; potential energy uses the cell ground plane as its height reference.

Limits and next experiment

Results apply to this small-carton model and its configured controller. They do not establish hardware safety, perception generalization, electrical consumption or industrial uptime. Camera semantic labels are simulator annotations; control uses simulator pose and mass.

Next: compare one physical condition at a time using matched seeds, retain all failures and evaluate the frozen choice on held-out starting positions. More repetitions are needed before making reliability claims.

Report generator: vivacity.factory-report/1.0. Missing or legacy provenance remains unavailable; it is never reconstructed as a claim about a past execution.